

MAE1001
Fall 2025

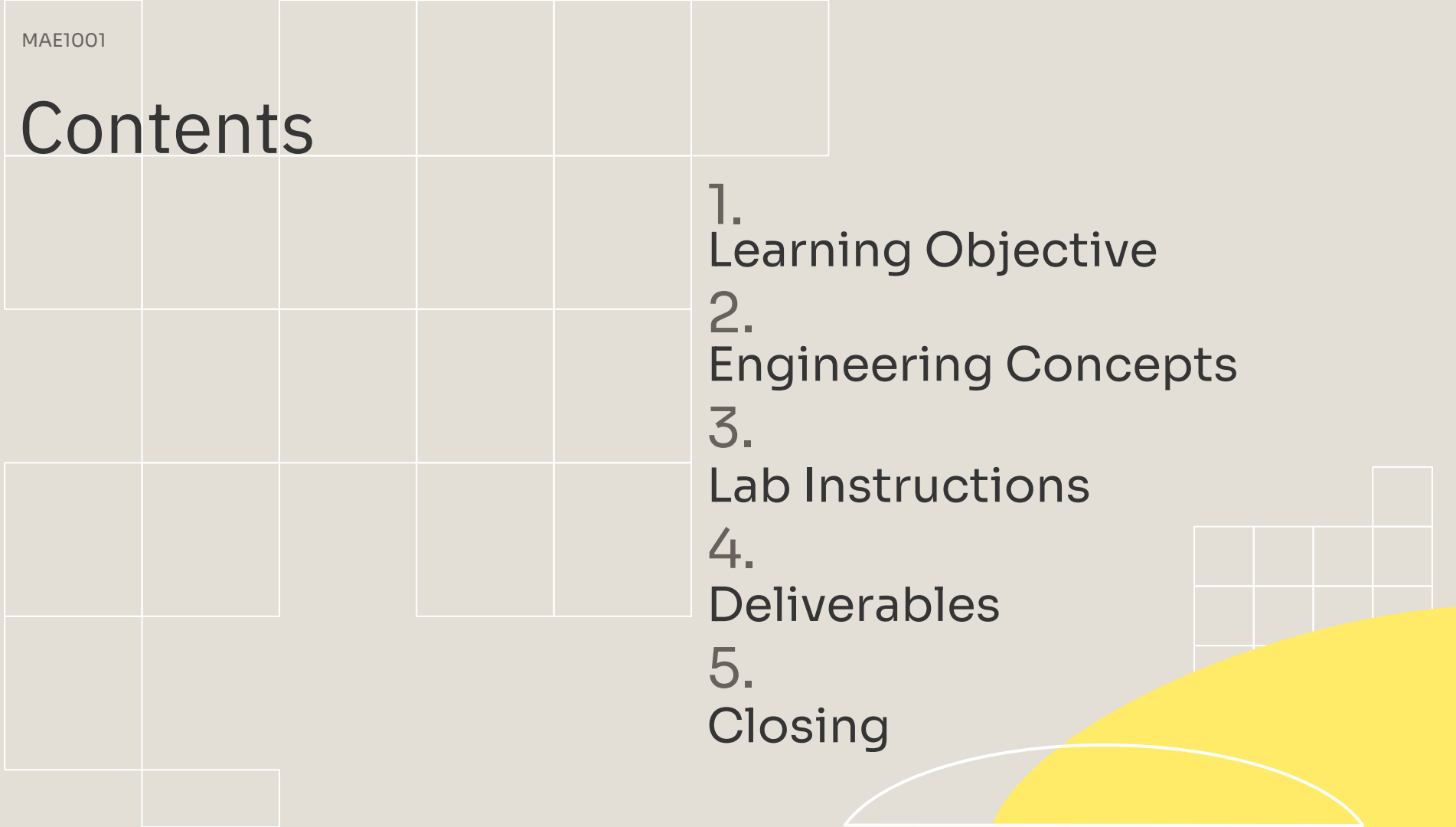
Screwdriver Disassembly

Presented by Shota Kakiuchi and Dr. Kartik Bulusu

Course webpage: <https://gwu-mae1001.github.io/fall2025/>



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- A decorative background featuring a light gray grid pattern on the left side. On the right side, there is a yellow curved shape at the bottom, resembling a rising sun or a stylized hill, with a white arc above it.

Learning Objective

Reverse Engineering

Take apart a screwdriver and identify how it works

Core Engineering Concepts

What are gears and motors, how does it create power (torque)

Ethical AI

Apply ethical AI to understand concepts and Equation.

Core Engineering Concepts I

Torque

Torque is rotational force

$$\tau = \text{Force} \times \text{distance}$$

Power

Power is the rate at which work is done or energy is transferred

$$\text{Power} = \text{Voltage} \times \text{Current}$$

Efficiency

How efficient the power is used, Output vs Input power

$$\text{Efficiency} = (\text{Output Power} / \text{Input Power}) \times 100$$

Gear Tooth

How many teeth are in the gear

$$n = \text{Tooth Number}$$

Gear Ratio

The relation of two meshing gears relative to Gear Teeth number.

$$\text{Gear Ratio} = \text{Driven} / \text{Drive}$$

Determines tradeoff between speed & torque

High gear ratio = more torque, less speed

Core Engineering Concepts II

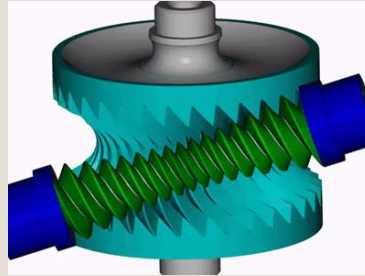
Types of Gears:

Spur Gear



Simple, cost-effective gears with straight teeth that transmit motion and power between two parallel shafts

Worm Gear



Screw-shaped gear (the worm) rotates and meshes with a toothed wheel (the worm wheel) at a 90 deg

Planetary Gear



Sun Gear in the Middle, Multiple Planetary Gears revolving around the Sun Gear, Ring Gear at the edge

Key Concepts

Discuss: How do the gears change torque vs. speed? Why are planetary gears preferred in screwdrivers?

1

DC Motor

Converts electrical energy to mechanical rotational energy

2

Planetary Gears

Multiplies torque while reducing speed and the opposite

3

Trigger Mech

Controls power input

4

Battery & Circuit Board

Supply and regulate electricity

Lab Instructions

Questions will be
asked for the Exit
Ticket

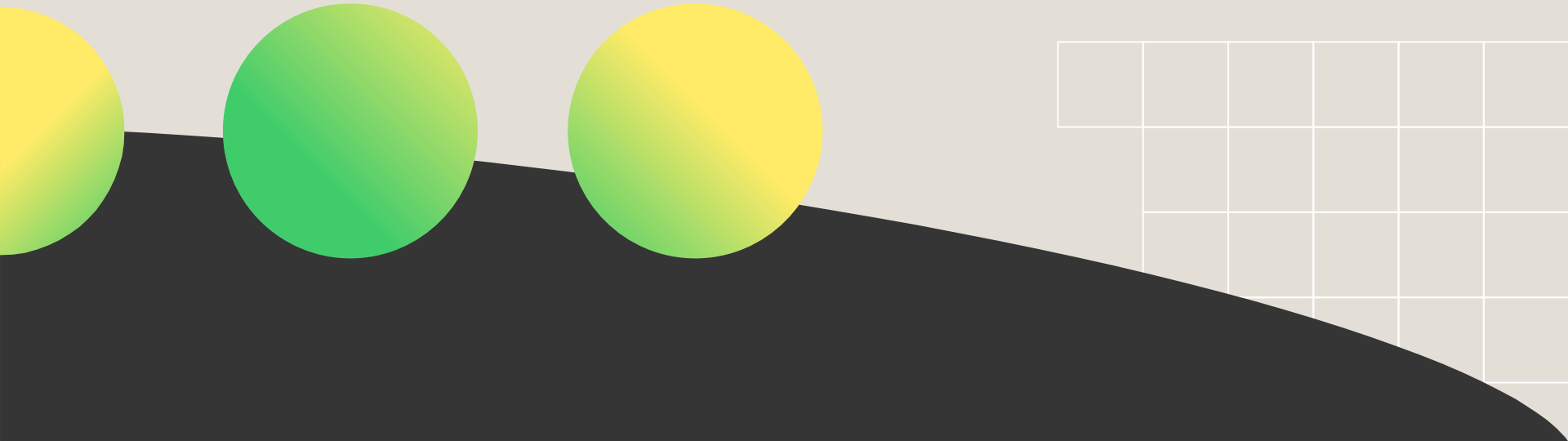
Materials Needed

Screwdriver per 2 students

Phone to take pictures of steps

Laptop to create lab notebook to record lab

For each step: Add Picture, Add Observations



Lab Instructions

Step

1

Remove
Battery Pack



Step

2

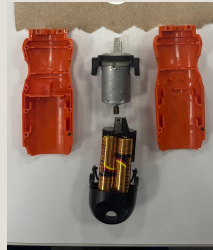
Remove Security
Clip and take the
two sections
apart



Step

3

Open the
orange casing
to reveal the
motor



Step

4

Remove all
components
from inside
head to reveal
gears



Goal



Now put it back
together



<- Motor Gear

Gear ->



1. How many Gear Teeth are on the Motor Gear and the Gear?
2. What is the Gear Ratio between the Motor Gear and Gear?
3. Use AI to Answer the following (Show Prompt and Answer):
 - a. Solve the Ring Gear Fixed **Gear Ratio** given the following teeth number, Sun Gear 10, Planet 20, Ring 50 ANS: 6:1

Questions

Add to Lab Notebook

Deliverables

11/21 Lab

We're here!

Complete the lab

12/04 Lab
Notebook

Submit a lab notebook that shows:
Each step taken, image of step, and
observations and description

12/04 Ethical AI

Answer the questions about Gears
using AI

MAE1001

Come to Office Hours

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Any questions? Ask away!



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